someone asked you to specify where the ant was at a given moment, you would need to give only one piece of data: the distance of the ant from the left (or the right) end of the hose. The upshot is that from a quarter of a mile away, a long piece of garden hose appears to be a one-dimensional object.

In reality, we know that the hose does have thickness. You might have trouble resolving this from a quarter mile, but by using a pair of binoculars you can zoom in on the hose and observe its girth directly, as shown in Figure 8.1(b). From this magnified perspective, you see that a little ant living on the hose actually has two independent directions in which it can walk: along the left-right dimension spanning the length of the hose as already identified, and along the "clockwise-counterclockwise dimension" around the circular part of the hose. You now realize that to specify where the tiny ant is at any given instant, you must actually give two pieces of data: where the ant is along the length of the hose, and where the ant is along its circular girth. This reflects the fact the surface of the garden hose is two-dimensional.1

Nonetheless, there is a clear difference between these two dimensions. The direction along the length of the hose is long, extended, and easily visible. The direction circling around the thickness of the hose is short, "curled up," and harder to see. To become aware of the circular dimension, you have to examine the hose with significantly greater precision.

This example underscores a subtle and important feature of spatial dimensions: they come in two varieties. They can be large, extended, and therefore directly manifest, or they can be small, curled up, and much more difficult to detect. Of course, in this example you did not have to exert a great deal of effort to reveal the "curled-up" dimension encircling the thickness of the hose. You merely had to use a pair of binoculars. However, if you had a very thin garden hose—as thin as a hair or a capillary—detecting its curled-up dimension would be more difficult.

In a paper he sent to Einstein in 1919, Kaluza made an astounding suggestion. He proposed that the spatial fabric of the universe might possess more than the three dimensions of common experience. The motivation for this radical thesis, as we will discuss shortly, was